

Nihal kumar singh

Hisua, Bihar 805103

+91 7479808459 nihalsinghwithroman@gmail.com [linkedin.com/in/nihal-kumar-singhdeveloper/](https://www.linkedin.com/in/nihal-kumar-singhdeveloper/)

github.com/nihalsingh571

Training

DSA - Summer Internship Training | Java, Javascript, Certification

May 2024 - Aug 2024

GeeksForGeeks.

- Completed a structured training program covering Data Structures and Algorithms from fundamentals to advanced concepts and solved 500+ DSA problems on GeeksForGeeks and LeetCode.
- Implemented real-world problem-solving techniques using Java with a strong focus on optimizing time and space complexity.
- Designed and developed an optimized Sudoku Solver using JavaScript and backtracking algorithms, applying recursion, constraint checking, and pruning to improve performance, with an interactive user interface built using HTML and CSS for real-time solution visualization.

Projects

InternRecom | React.js, TF-IDF, Cosine Similarity, Ansible, Terraform, Nagios, Docker, AWS

Oct 2025 - Jan 2026

- Designed and implemented a backend recommendation engine using Node.js and Express.js to serve internship recommendations.
- Implemented a TF-IDF + cosine similarity-based text matching model for internship recommendations and designed a normalized [0–1] scoring pipeline combining cosine similarity, VSPS, and Trust Score.
- Provisioned infrastructure using Terraform (IaC) and automated configuration with Ansible, Containerized applications using Docker, deployed on AWS, and monitored system health using Nagios.

E-Commerce | React.js, MongoDB, Cloudinary, Jenkins, Docker, AWS

Apr 2025 - May 2025

- Developed a fully functional e-commerce platform with multiple core features.
- Implemented user authentication, product availability checks, and cart management functionality, Verified product availability across 1000+ pin codes to ensure accurate delivery feasibility.
- Dockerized the application using Docker images and containers for consistent deployment, Automated CI/CD pipelines using Jenkins and GitHub webhooks, reducing deployment time by 70%.

Secure RESTful Api | Node.js, Express, MongoDB, JWT, and Postman

Jan 2025 - Feb 2025

- Implemented JWT-based authentication supporting two user roles (admin and user).
- Applied express-validator for input validation and Helmet for securing HTTP headers, Implemented pagination and filtering, reducing API load time by 35%.
- Integrated Swagger for clear, interactive, and maintainable API documentation.

Certificates / Certifications

Java Programming - GFG

Sep 2025

Cloud Computing - NPTEL

Oct 2024

DSA- GFG

SEP 2024

Skills

Languages: Java, Python, JavaScript

Technologies/Frameworks: React JS, Node Js, Django

Domain Skills: DSA, OOPS, DevOps, OS, CN, Problem-Solving, Responsive Web Design, Scripting in Python and JavaScript

Tools: MySQL, MongoDB, Git, Docker, AWS, Linux, Jenkins, Canva, Puppet, Nagios, Terraform, Ansible

Education

Lovely Professional University

2022 – 2026

Bachelor of Technology - Computer Science and Engineering — CGPA: 7.53

Phagwara, Punjab

Jeevan Jyoti Public School

2019 – 2021

12th — Percentage: 73.6%

Nawada, Bihar

Jeevan Jyoti Public School

2018 – 2019

10th — Percentage: 91.2%

Nawada, Bihar